25-10-21

Lecture - 09
chapter - 03

Functions of one random variables and expectations

X be the discrete random variable, the the expected value of X.

$$E[X] = \sum_{i=1}^{\text{limit}} x_i f(x_i); \quad 0 < f(x_i) < 1$$

In case of X, be continuous random variable,

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$E(aX+b) = aE[x] + b$

Ex: if $f(x) = \frac{1}{\beta - \alpha}$; $\alpha < x < \beta$, x – CRV

$$E[X] = \int_{\alpha}^{\beta} x \frac{1}{\beta - \alpha} dx$$

$$= \frac{1}{\beta - \alpha} \left[\frac{x^2}{2}\right]_{\alpha}^{\beta}$$

$$= \frac{1}{2(\beta - \alpha)} (\beta^2 - \alpha^2)$$

$$= \frac{\beta + \alpha}{2}$$

Scanned with CamScanner

X - be a random variable and $g(x)$ be a function of random variable. then

$$E[g(x)] = \begin{cases} \sum_{i=1}^{n} g(x_i) f(x_i) & - \text{x DRV} \\ \int_{-\infty}^{\infty} g(x) f(x)\, dx & - \text{x CRV} \end{cases}$$

Ex:

$g(x) = x^2$, $f(x) = \frac{1}{9}(4x - x^2)$; $1 < x < 4$

$$E[g(x)] = \int_1^4 x^2 \frac{1}{9}(4x - x^2)\, dx$$

$$= \frac{1}{9}\int_1^4 (4x^3 - x^4)\, dx$$

$$= \frac{4}{9}\left[\frac{x^4}{4}\right]_1^4 - \frac{1}{9}\left[\frac{x^5}{5}\right]_1^4$$

$$= \frac{1}{9}(4^4 - 1^4) - \frac{1}{45}(4^5 - 1^5)$$

$$= 5.6$$

Properties -

1) $E[ax] = aE[x]$ [a constant]

2) $E[ax+b] = aE[x] + b$ [a, b "]

3) $E[g(x) \pm h(x)] = E[g(x)] \pm E[h(x)]$

4) $E[g(x_1) + g(x_2) + \cdots + g(x_n)] = E[g(x)] + \cdots$

Scanned with CamScanner

Median

$$\int_{\alpha}^{M} f(x)\,dx = \frac{1}{2} = \int_{M}^{\infty} f(x)\,dx$$

Variance:

X - RV with mean μ

$$V[x] = E[X - E(x)]^2$$

$$= E[X - \mu]^2$$

$$= \begin{cases} \sum (x_i - \mu)^2 f(x) \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx \end{cases}$$

$$V[x] = E[X^2] - (E[X])^2$$

standard Dev, $\Delta = \sqrt{V(X)}$

$V(aX+b) = a^2 V(X)$

$V[ax] = a^2 V(x)$

$V[b] = 0$

$V[x] = V[-x]$

$V[x \pm Y] = V[x] + V[Y] \pm 2\,cov(xY)$

$V[Ax + BY] = A^2 V[x] + B^2 VY \pm 2AB\,cov(xY)$

if x, Y independent. $cov(xY) = 0$

Scanned with CamScanner

Moment generating function:

Given a random variable X, the moment-generating function $M_X(t)$ of its probability distribution is the expected value of e^{tx}.

$$M_X(t) = E(e^{tx})$$

$$= \begin{cases} \sum_i e^{tx_i} f(x_i) & x \text{ discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\, dx & x \text{ continuous} \end{cases}$$

Ex 3.11,

Scanned with CamScanner

3-19. The repair time (in hours) for a certain electronically controlled milling machine follows the density function

$$f_X(x) = 4xe^{-2x}, \qquad x > 0,$$
$$= 0, \qquad \text{otherwise.}$$

Determine the moment-generating function for X and use this function to evaluate $E(X)$ and $V(X)$.

Scanned with CamScanner

3.19) $f_x(x) = 4xe^{-2x} \quad x > 0$

$$m_x(t) = \int_0^\infty e^{tx} 4xe^{-2x} \, dx$$

$$= 4\int_0^\infty x e^{tx-2x} \, dx$$

$$= 4\left\{ \left[x \frac{e^{-x(2-t)}}{-(2-t)} \right]_0^\infty - \int \frac{e^{-x(2-t)}}{-(2-t)} dx \right.$$

$$= \frac{4}{2-t} \left[\frac{e^{-x(2-t)}}{-(2-t)} \right]_0^\infty$$

$$= \frac{-4}{(2-t)^2} \{ 0 - 1 \)$$

$$= \frac{4}{(2-t)^2}$$

$$= \frac{4}{4(1-\frac{t}{2})^2}$$

$$= (1-\tfrac{t}{2})^{-2}$$

Scanned with CamScanner

Mean

$$\int_0^\infty 4x^2 e^{-2x}\, dx$$

$$= 4\left[\frac{x^2 e^{-2x}}{-2}\right]_0^\infty - 4\int \frac{2xe^{-2x}}{-2}\, dx$$

$$= 4\int xe^{-2x} dx$$

$$= 4\left\{\left[\frac{xe^{-2x}}{-2}\right]_0^\infty - \int_0^\infty \frac{e^{-2x}}{-2}\, dx\right\}$$

$$= 2\int_0^\infty e^{-2x}\, dx$$

$$= -1\left(e^{-2x}\right)_0^\infty$$

$$= -1(0-1)$$

$$= 1$$

Scanned with CamScanner

Variance

$$V(x) = E(x^2) - E(x)^2$$

$$E(x^2) = \int_0^\infty x^2 \, 4xe^{-2x} \, dx$$

$$= 4\int_0^\infty x^3 e^{-2x} \, dx$$

$$= 4\left\{\left[\frac{x^3 e^{-2x}}{-2}\right]_0^\infty - \int_0^\infty 3x^2 \frac{e^{-2x}}{-2} \, dx\right\}$$

$$= 4 \times \frac{3}{2} \int_0^\infty x^2 e^{-2x} \, dx$$

$$= 6\left[\frac{x^2 e^{-2x}}{-2}\right]_0^\infty - 6\int \frac{2xe^{-2x}}{-2} \, dx$$

$$= 6\int_0^\infty xe^{-2x} \, dx$$

$$= 6\left[\frac{xe^{-2x}}{-2}\right]_0^\infty - 6\int_0^\infty \frac{e^{-2x}}{-2} \, dx$$

Scanned with CamScanner

$$= 3\int_0^\infty e^{-2x}\,dx$$

$$= 3\,\frac{-1}{2}\,(e^{-2x})_0^\infty$$

$$= \frac{3}{2}$$

$$V(x) = \frac{3}{2} - 1$$

$$= \frac{1}{2}.$$

—o—

Scanned with CamScanner

Example 3-11

A well-known simple inventory problem is "the newsboy problem," described as follows. A newsboy buys papers for 15 cents each and sells them for 25 cents each, and he cannot return unsold papers. Daily demand has the following distribution and each day's demand is independent of the previous day's demand.

Number of customers x	23	24	25	26	27	28	29	30
Probability, $p_X(x)$	0.01	0.04	0.10	0.10	0.25	0.25	0.15	0.10

If the newsboy stocks too many papers, he suffers a loss attributable to the excess supply. If he stocks too few papers, he loses profit because of the excess demand. It seems reasonable for the newsboy to

Scanned with CamScanner

stock some number of papers so as to minimize the *expected* loss. If we let s represent the number of papers stocked, X the daily demand, and $L(X, s)$ the newsboy's loss for a particular stock level s, then the loss is simply

$$\begin{aligned} L(X, s) &= 0.10(X - s) && \text{if } X > s, \\ &= 0.15(s - X) && \text{if } X \le s, \end{aligned}$$

and for a given stock level, s, the expected loss is

$$E[L(X,s)] = \sum_{x=23}^{s} 0.15(s-x)\cdot p_X(x) + \sum_{x=s+1}^{30} 0.10(x-s)\cdot p_X(x)$$

and the $E[L(X, s)]$ is evaluated for some different values of s.

For $s = 26$,

$$\begin{aligned} E[L(X, 26)] &= 0.15[(26 - 23)(0.01) + (26 - 24)(0.04) + (26 - 25)(0.10) \\ &\quad + (26 - 26)(0.10)] + 0.10[(27 - 26)(0.25) + (28 - 26)(0.25) \\ &\quad + (29 - 26)(0.15) + (30 - 26)(0.10)] \\ &= \$0.1915. \end{aligned}$$

For $s = 27$,

$$\begin{aligned} E[L(X, 27)] &= 0.15[(27 - 23)(0.01) + (27 - 24)(0.04) + (27 - 25)(0.10) \\ &\quad + (27 - 26)(0.10) + (27 - 27)(0.25)] + 0.10[(28 - 27)(0.25) \\ &\quad + (29 - 27)(0.15) + (30 - 27)(0.10)] \\ &= \$0.1540. \end{aligned}$$

For $s = 28$,

$$\begin{aligned} E[L(X, 28)] &= 0.15[(28 - 23)(0.01) + (28 - 24)(0.04) + (28 - 25)(0.10) \\ &\quad + (28 - 26)(0.10) + (28 - 27)(0.25) + (28 - 28)(0.25)] \\ &\quad + 0.10[(29 - 28)(0.15) + (30 - 28)(0.10)] \\ &= \$0.1790 \end{aligned}$$

Thus, the newsboy's policy should be to stock 27 papers if he desires to minimize his expected loss.

Scanned with CamScanner